Yu Chul (Jerry) Lee

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Education

PhD **The University of Texas at Austin**, Mechanical Engineering — Jan 2025 – May 2028
- Expected graduation: May 2028

MS **The University of Texas at Austin**, Mechanical Engineering — Aug 2023 – Dec 2024
- GPA: 4.0/4.0

BS **Yonsei University**, Mechanical Engineering — Mar 2019 – Aug 2023
- GPA: 3.71/4.5

Research Experience

Mobility Systems Laboratory, UT Austin, Graduate Research Assistant — Jan 2025 – present
- Developed hybrid physics-informed machine learning architectures for driver behavior prediction at four-way intersections.
- Designed interaction-aware and driver-mode-aware Intelligent Driver Models augmented with learned latent representations.
- Trained and evaluated models using 163 NGSIM intersection scenarios and human-driver simulator datasets.
- Achieved 0.623 m/s velocity RMSE over a 2-second prediction horizon.
- Reduced unseen-scenario acceleration prediction RMSE from 20.5 m/s^2 to 4.06 m/s^2 compared to EBM baselines.
- Developed MATLAB and Python pipelines for scenario extraction, feature engineering, training, and rollout analysis.

Mobility Systems Laboratory, UT Austin, Researcher – Explainable Driver Modeling — Jan 2024 – present
- Developed interpretable driver models using SHAP-based feature selection and Explainable Boosting Machines.
- Engineered physically meaningful driver-behavior features from vehicle kinematics and traffic interactions.
- Proposed residual-learning adaptation methods for personalized driver modeling from sparse datasets.
- Achieved 94.4% fidelity for synthetic drivers and up to 88.5% fidelity for human-driver behavior.
- Preserved individual driving styles and achieved 75% driver identification accuracy on unseen data.

Engineering Experience

EcoCAR Connected and Autonomous Vehicle Team, Team Leader — Jan 2025 – May 2026
- Led a multidisciplinary team of 8 undergraduate and graduate students developing CAV technologies on a Cadillac LYRIQ.
- Directed development of CACC, LKA, Automated Intersection Navigation, sensor fusion, and V2X-enabled autonomy features.
- Coordinated model-based development workflows using MATLAB/Simulink, RTMaps, dSPACE ASM, and dSPACE AUTERA.

EcoCAR, Lead Developer – Cooperative Adaptive Cruise Control (CACC) Jan 2025 – May 2025

- Co-developed an MPC-based CACC system utilizing SAE J2735 V2V communication.
- Integrated V2X vehicle state information into real-time control algorithms.
- Implemented CAN-FD communication between dSPACE AUTERA and vehicle control systems.
- Successfully deployed and validated the controller on a Cadillac LYRIQ.
- Achieved 0.062 m steady-state spacing RMSE, 2.999 m/s^3 maximum jerk, and 1.707 m/s^2 maximum acceleration.

EcoCAR, Lead Developer – Lane Keeping Assist (LKA) Jan 2025 – May 2025

- Guided development of a Sliding Mode Controller using LiDAR and camera perception inputs.
- Mentored undergraduate team members in controller design and validation.
- Validated controller performance through HIL and vehicle testing.
- Achieved 0.121 m lateral tracking error and 0.35 m/s^3 lateral jerk in HIL testing.
- Conducted vehicle-in-the-loop testing and identified sensor latency limitations.

Selected Projects

The University of Texas at Austin, Project Lead – Autonomous Vehicle Path Tracking Jan 2024 – Dec 2024

- Designed and tested a path-tracking controller capable of lane changes at 40 m/s with 0.1 m lateral error.
- Used root-locus and frequency-domain design methods to satisfy system requirements.
- Reduced lateral tracking RMSE by 52.98% for single lane changes and 48.61% for double lane changes.
- Achieved performance comparable to MPC with significantly lower computational cost.

Yonsei University, Team Member – Advanced Mechatronics Competition (1st Place) Aug 2023 – Dec 2023

- Designed and manufactured an autonomous mobile robot capable of target localization and pneumatic actuation.
- Created 12 custom SolidWorks components under strict dimensional constraints.
- Tested five prototypes and achieved reliable 1.5 m projectile launch capability.

Publications

Practical Method for High-Speed Path-Tracking Control of Vehicles with Time-Delayed Steering System Dynamics Jan 2025
Yu Chul Lee, Chi-Chang Huang, Umesh Malepati, Xingyu Zhou, Junmin Wang
(IEEJ Journal of Industry Applications)

Human Impact of Visual Detail in Vehicle Lane-Keeping System Communication May 2025
Yu Chul Lee, Junmin Wang
(IFAC Symposium on Intelligent Autonomous Vehicles)

Technologies & Skills

Controls & Vehicle Dynamics: Model Predictive Control (MPC), Sliding Mode Control, LQR, Feedback Linearization

Programming: MATLAB, Simulink, Python, C

Autonomous Systems: V2X Communication (SAE J2735), CAN-FD, Driver Modeling, Trajectory Prediction

Simulation & Deployment: dSPACE AUTERA, dSPACE ASM, RTMaps, Unreal Engine, ROS, CANalyzer, HIL, VIL

Machine Learning & Data Analysis: Explainable Boosting Machines (EBM), SHAP, Random Forests, Clustering

Certifications

Professional Certification: Fundamentals of Engineering (FE Mechanical) — Passed Feb 2023